PROVEIT RESEARCH REPORT

Shared Binomial Certificates and Short Carriers

Explicit Diophantine forms of Con(ZFC)

Prepared for Vladimir Reshetnikov

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Main outputs. A direct quartic in 319 nonnegative integers; an eleven-integer equation of degree 966 with a 111-operation evaluator; and an eleven-integer equation of degree 440 with a 126-operation evaluator. Every coefficient and arithmetic operation is supplied in the companion archive. The new constructions preserve the finite-refutation endpoint of the explicitly retained ordinary source proofs.

Main idea. Reuse the same binomial certificate to certify a factorial and the product of the CRT moduli. On the small-variable side, move a coprime square condition into a Pell coefficient, and use a short carrier whose weaker divisibility is sufficient for exact recovery.

Scope. These are concrete improvements against the inspected repository and all eight incoming archives. No global minimum, historical priority theorem, proof of consistency, or newly completed Lean/Rocq endpoint is claimed. The arithmetic proofs, finite artifact checks, and inherited semantic calibration are identified separately.

Comparison pin:

`45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`

https://github.com/VladimirReshetnikov/ProveIt

Abstract

We construct and export new fixed polynomial equations whose absence of solutions is equivalent, by ordinary arguments formalizable in ZFC and the retained source calibration, to Con(ZFC). The direct proof-certificate route has 319 nonnegative variables, degree four, 254 quadratic residuals, 10,283 collected monomials, and coefficient height 86,842,335. It improves the inspected 349-variable endpoint in variables, residuals, terms, and height. Its architecture needs only six full power certificates instead of nineteen: a single exact binomial serves both the factorial and the reciprocal CRT product, while common Pell residues handle eleven digit requests. This route does not simulate K215 or specialize Jones's universal polynomial.

A complementary native-source construction uses eleven unrestricted integer variables. A coprime product is absorbed into the strong Pell coefficient; congruences modulo four recover positivity of the extracted factors. Together with a short powered-divisibility carrier this gives degree 966 at 111 operations, or degree 882 at 112 operations. A three-root companion gives degree 464 at 125 operations and degree 440 at 126 operations. The report supplies the mathematical definitions, both directions of the new reductions, the exact degree arguments, the coefficient conventions, and a reproducible audit. Existing lemmas and unbounded source theorems are credited explicitly.

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1 Outcome and the meaning of simplicity

The target is a *fixed* polynomial P with integer coefficients and a specified domain D such that

$$\mathsf{ZFC} \vdash \left(\neg\exists \mathbf{x} \in D^m \; P(\mathbf{x}) = 0\right) \iff \mathrm{Con}(\mathsf{ZFC}). \tag{1}$$

Here $\mathbb{N} = \{0, 1, 2, \ldots\}$; $\mathbb{Z}$ means unrestricted integers. A named polynomial intermediate is not an additional quantified variable. The fixed coefficients are literal finite data, reconstructed without running a refutation search. The equivalence does not assume consistency.

There is no single agreed numerical order on all such equations. Variable count, degree, collected support, coefficient height, and arithmetic circuit length measure different costs. An enormous integer hidden in a coefficient is not free from an overall description-length perspective. Accordingly we give a small set of explicitly measured tradeoffs.

1.1 Measured improvements

For the first table, operations count every live binary addition, subtraction, and multiplication, including multiplication by a fixed scalar and squaring. Reuse of an existing register is free. Fixed coefficient leaves are reported separately.

Construction over $\mathbb{Z}$	Variables	Degree	Operations	Status
Incoming coprime fusion, direct	11	728	122	baseline
Incoming coprime fusion, centered	11	688	126	baseline
Incoming all-root fusion	11	1632	114	baseline
Short carrier, direct	11	488	121	this report
Short carrier, centered	11	464	125	this report
Short carrier, smaller index modulus	11	440	126	this report
Pell-coefficient saturation	11	966	111	this report
Saturation, smaller index modulus	11	882	112	this report

The 464/125 point improves both degree and cost relative to 688/126. The 440/126 point reduces the same baseline degree to 440 at unchanged cost. The 966/111 point improves both measures relative to 1632/114. These statements compare the same unrestricted integer domain and the same six fixed coefficient values.

For a collected polynomial $\sum_\alpha c_\alpha X^\alpha$, its height is $\max_\alpha |c_\alpha|$, and its support size counts nonzero collected coefficients. Both following equations are quartics over $\mathbb{N}$.

Construction	Variables	Residuals	Terms	Height
Incoming factorial-carrier endpoint	349	268	12,768	86,915,503
Shared binomial and modular residues	319	254	10,283	86,842,335

This removes 30 variables (8.60%), 14 residuals, and 2,485 terms (19.46%). The number of full power certificates drops from 19 to 6; the number of exact binomial certificates drops from two to one. All coefficients of both expanded quartics were independently re-expanded from their residual lists for the comparison.

1.2 The baseline includes the incoming archives

The comparison uses the public repository at the pin on the title page, including its latest proposal review, the active candidate and research directories, and all eight ZIPs then in `docs/incoming`. The

newer incoming reports dominate several entries of the main comparison table. In particular, the 349-variable quartic and the eleven-integer 688/126 and 1632/114 circuits are incoming baselines, not results of this work. Their source paths and hashes are retained in the archive provenance.

Other architectures remain relevant. The repository already has nine-positive-variable equations and a ten-integer construction, and it retains a 37-integer quartic with enormous fixed coefficients. The new 319-variable quartic is therefore not a fewest-variable quartic record. Its advantage is within a direct, explicitly expanded proof-certificate architecture with modest numerical coefficients. The eleven-variable native-source equations instead use six very large fixed integers.

1.3 Attribution and research scope

The squared-radius factorial identity used below was already proved in package 105 and appears in the incoming synchronized-exponents work. Common modular Pell residues occur in package 108. The reciprocal CRT product, derived label range, and shared-factorial Horn-flow source occur in Round 14. The new direct contribution is their complete integration: the factorial and CRT binomials become *the same certificate*, the digit base is validated before that certificate is interpreted, and a complete smaller quartic is exported.

On the small-variable side, powered divisibility and supported-modulus Pell interpolation come from Round 14, and selective coprime fusion comes from the incoming coprime-projection report. The new work combines them with a smaller index modulus and a Pell-coefficient saturation that recovers signs by congruences modulo four. This report makes no claim that these ideas are absent from all mathematical literature.

2 Logical endpoint and evidence boundaries

The common semantic endpoint is the existence of a finite syntactic refutation. For a fixed, effective presentation of ZFC, write

$$\mathrm{Ref}_{\mathsf{ZFC}} \equiv \exists p\ \mathrm{Proof}_{\mathsf{ZFC}}(p, \bot), \qquad \mathrm{Con}(\mathsf{ZFC}) \equiv \neg\,\mathrm{Ref}_{\mathsf{ZFC}}\,.$$

Each construction proves $\exists P = 0 \leftrightarrow \mathrm{Ref}_{\mathsf{ZFC}}$. Negation gives (1). Nothing in this argument proves $\mathrm{Con}(\mathsf{ZFC})$ itself or decides whether a zero exists.

Theorem 2.1 (Ordinary correctness of the delivered reductions)**.** *Compose the explicitly retained ordinary native/calculus-to-refutation theorems with the arithmetic reductions proved in this report. The polynomial in* `direct/equation319/natural/equation.json` *has a zero in* $\mathbb{N}^{319}$ *if and only if ZFC has a finite refutation. Each of the five selected eleven-variable polynomials in the comparison table has a zero in* $\mathbb{Z}^{11}$ *if and only if ZFC has a finite refutation. These finite arithmetical arguments are formalizable in ZFC.*

Three kinds of evidence must be kept separate.

1. **Ordinary mathematical proofs.** The report and retained sources establish the unbounded number-theoretic and refutation equivalences, including witness existence and all sign cases.
2. **Finite exact checks.** Independent programs authenticate coefficient bytes, reconstruct the fixed tuple, verify the entire polynomial expressions and elimination certificates, count operations, and certify degrees. These checks do not decide consistency.
3. **Proof-assistant status.** ProveIt has separate formal endpoints and extensive local proofs. This archive supplies no new end-to-end Lean/Rocq build and no literal internal object-language ZFC

derivation of the newly displayed equations. Ordinary formalizability is the claim in this report; a new kernel certificate is a further task.

The direct route starts from the numerical C222 calculus and its retained ordinary equivalence to finite ZF refutability, followed by the retained ZF/ZFC refutation equivalence. The statement is applied before a machine-specific simulation endpoint: the new direct builder does not read or simulate the K215 transition table. The small-variable branch deliberately retains the complete existing native-source coefficient tuple and its calibration.

3 One binomial certificate for factorial and CRT arithmetic

The strongest direct result of this investigation is a literal quartic in 319 nonnegative integers. It improves the latest inspected direct natural equation in four measured quantities at once:

	variables	quadratic rows	monomials	height
Round 14 derived-label source	349	268	12 768	86 915 503
Merged factorial–CRT source	319	254	10 283	86 842 335.

Both single equations have exact degree four. The comparison concerns direct C222 equations with completely collected coefficients. It does not assert superiority to the repository's few-variable equations with different coefficient and degree costs.

Theorem 3.1 (Explicit direct equation)**.** *Let $R_1, \dots, R_{254} \in \mathbb{Z}[z_0, \dots, z_{318}]$ be the complete quadratic residual list in the delivered file* `direct/equation319/natural/equation.json`*, and put*

$$P_{319}(z) = \sum_{j=1}^{254} R_j(z)^2.$$

The supplied collected coefficient file defines the same polynomial. Its degree is four, it has 10 283 *nonzero monomials, and its height is* 86 842 335*. With the inherited numerical-calculus calibration,*

$$\mathrm{ZFC} \vdash (\nexists z \in \mathbb{N}^{319} : P_{319}(z) = 0) \iff \mathrm{Con}(\mathrm{ZFC}).$$

Here $\mathbb{N} = \{0, 1, 2, \dots\}$; all coefficients are fixed, literal integers.

The ordinary proof is given below. Its arithmetic source has 465 natural coordinates and 400 quadratic rows; an independent exact certificate eliminates 146 coordinates, including a nonnegativity proof for every restoration. Expansion of all remaining squares verifies every final coefficient. These finite checks support the literal construction; the unbounded equivalence is proved mathematically, rather than inferred from tests. No new end-to-end Lean proof term or emitted object-language ZFC derivation is claimed.

3.1 The improvement and its mathematical provenance

The preceding reciprocal-CRT construction used $F = t!$ and

$$B = \binom{F^2 + t}{t},$$

but obtained the factorial through a different binomial, $\binom{Ft^2+t}{t}$. We align these two uses of binomial arithmetic. The squared-radius factorial fixed-point theorem was already established in package 105,

Bootstrap certificates, Section 2 of its `NEW_RESULTS.md`; it is not a new theorem of this iteration. The common Pell-residue representation was already established in package 108, *Pell residues instead of power values*. The shared-factorial source, reciprocal prime-power CRT and derived-label range come from the latest Round 14 packet. These source proofs are retained in the archive.

The completed contribution is their integration: the *same* binomial output serves the factorial and CRT product; an independent radix bound validates that output before either semantic interpretation is assumed; common Pell residues eliminate eleven full numerator powers; and the entire construction is compiled, contracted and measured. This is a new complete equation relative to the inspected snapshot, without a claim of worldwide priority for the ingredients.

3.2 The inherited factorial identity, with an independent proof

Lemma 3.2 (Squared-radius factorial)**.** *For $t \geq 6$ and positive integer F,*

$$F = \left\lceil \frac{(F^2+1)^t}{\binom{F^2+t}{t}} \right\rceil \quad \Longleftrightarrow \quad F = t!.$$

Proof. Put

$$H_t(F) = t! \prod_{i=1}^{t} \frac{F^2+1}{F^2+i}.$$

The ceiling assumption gives $H_t(F) \leq F \leq t!$. If $F < t$, monotonicity of each factor in its radius gives

$$H_t(F) \geq \frac{2^t}{t+1} > t > F,$$

a contradiction. The middle inequality follows at $t = 6$ and persists by induction. Thus $F \geq t$.

Using $\prod(1-u_i) \geq 1 - \sum u_i$,

$$\prod_{i=1}^{t} \frac{F^2+1}{F^2+i} > 1 - \frac{t(t-1)}{2F^2} > \frac{1}{2}.$$

If $F < t^2$, then $H_t(F) > t!/2 > t^2 > F$, again impossible. Here $t! > 2t^2$ for $t \geq 6$ follows immediately by induction. Hence $F \geq t^2$, and the stronger estimate is

$$H_t(F) > t! \left(1 - \frac{1}{2F}\right).$$

If $t! \geq F+1$, the right side is at least $(F+1)(1-1/(2F)) = F + 1/2 - 1/(2F) \geq F$, contradicting $H_t(F) \leq F$. Therefore $F = t!$. Conversely, for $F = t!$, the same estimate gives $F - 1/2 < H_t(F) \leq F$, so its ceiling is F. □

Package 105 proves the more general statement for every nonnegative input. The restricted proof above is enough here because the horizon is padded to at least 16. The ceiling has an exact natural graph:

$$V + B = BF + r + 1, \qquad B = r + s + 1, \qquad r, s \in \mathbb{N}. \tag{2}$$

For $B > 0$ these rows give $V + s = BF$ and $0 \leq s < B$, hence $F = \lceil V/B \rceil$. If $V \geq 1$ they also force $F \geq 1$.

3.3 A radix that can be justified before the factorial

Define, using natural variables t_0, x, y, F, b_0,

$$\begin{aligned} t &= t_0 + 16, \qquad R = 2 + t + x + y + F, \qquad K = 2R, \\ B &= b_0 + 2, \qquad N = B^K, \qquad q = N^K, \qquad V = (F^2+1)^t. \end{aligned} \tag{3}$$

These variable powers are explanatory requests: their final implementation uses the explicit indexed-Pell graph below. Once they have been recovered, the coordinate shift $B \geq 2$ gives

$$N \geq 2^K, \qquad q = B^{K^2} \geq 2^{4R^2}. \tag{4}$$

No interpretation of B as a binomial is needed for this bound.

For $n = F^2 + t$ and $0 \leq j \leq t$,

$$\binom{n}{j} \leq n^t \leq (R^2+R)^t \leq 2^{2Rt} < 2^{2R^2} < q. \tag{5}$$

Use the zero convention if $j > n$. The elementary bound $R^2 + R \leq 4^R$ holds for $R \geq 1$, and $t < R$. Thus the first exact digit graph can be validated *before* any factorial or CRT meaning is available. It yields $B = \binom{F^2+t}{t}$; equations (2) and Lemma 3.2 then yield $F = t!$.

This order removes the potential circularity in identifying the two binomials. The previous radix $N(N+V+1)^K$ is replaced by N^K; its bootstrap is supplied by an explicit lower bound on B, not by assuming the value of the very digit being tested.

3.4 Six exact powers and eleven modular Pell residues

For $a \geq 2$, write

$$X_0(a) = 1, \quad Y_0(a) = 0, \quad X_{j+1} = aX_j + (a^2-1)Y_j, \quad Y_{j+1} = X_j + aY_j.$$

The retained compiler supplies indexed-Pell certificates for these pairs, sharing an auxiliary Pell solution and CRT base across all requested indices. The final system uses exactly six full power requests:

$$\begin{aligned} V &= (F^2+1)^t, \quad N = B^K, \qquad q = N^K, \\ D_t &= q^t, \qquad\qquad D_y = q^{y+1}, \quad D_x = q^{x+1}. \end{aligned} \tag{6}$$

The denominator $D_{t+1} = qD_t$ is obtained by multiplication, so it needs no separate power request or Pell pair at index $t+1$.

Let W be two plus the sum of the distinct bases, exponents and outputs in (6). The common frame is

$$L^2 - ((W+1)^2 - 1)(WZ)^2 = 1, \qquad Z \geq 1, \qquad A = L + \sigma. \tag{7}$$

All coordinates are natural. Pell classification and $Y_e(W+1) \equiv e \pmod{W}$ imply $L > W^W$, hence $A > W^W$. Conversely the index $e = W$ supplies L, Z, after which every $A \geq L$ is allowed.

For an exact request $m = n^k$, the compiler asserts

$$T_n = 2An - n^2 - 1 \geq 0, \qquad X_k(A) + nY_k(A) = AY_k(A) + m + T_n h, \quad h \geq 0.$$

The first row excludes $n = 0$. The frame bounds m and n^k strictly below T_n, and the Pell recurrence gives their congruence modulo T_n, so $m = n^k$. Every exponent is positive by its literal definition. This

is the inherited simultaneous positive-power theorem; separating the lower bound L from the actual base A permits the congruence below.

Put

$$b = q + 1, \qquad \mathcal{M} = qD_{t+1}D_xD_y = q^{t+x+y+4}, \tag{8}$$

and impose one natural-quotient equation

$$2A(q+1) = (q+1)^2 + 1 + \mathcal{M}h_0. \tag{9}$$

For every $n \geq 0$ it yields

$$\mathcal{R}_n = X_n(A) + (q + 1 - A)Y_n(A) \equiv (q+1)^n \pmod{\mathcal{M}}. \tag{10}$$

Indeed the substitution $s = q + 1 - A$ respects $s^2 = A^2 - 1$ modulo $\mathcal{M}$, and $(A + s)^n = X_n + Y_n s$. Moreover $\mathcal{R}_n > 0$: for $n > 0$, $X_n > (A - 1)Y_n$, and for $n = 0$ the value is 1.

For every required denominator d, both qd and Nd divide $\mathcal{M}$: the first follows from its product definition and the second from $N \mid q$. The one exact binomial uses

$$\begin{aligned} X_n + (q+1)Y_n &= AY_n + r + dB + qdh, \\ d &= r + s + 1, \qquad\qquad q = B + o + 1, \end{aligned} \tag{11}$$

where $n = F^2 + t$, $d = D_t$. Every divisibility request $N \mid \binom{n}{k}$ instead uses

$$X_n + (q+1)Y_n = AY_n + r + Ndh, \qquad d = r + s + 1, \qquad d = q^k. \tag{12}$$

The binomial theorem identifies the digit or its divisibility once all lower coefficients are below q. Positivity of $\mathcal{R}_n$ ensures that the constructed quotients are nonnegative.

Completeness of the common root is not automatic and must be checked. As $\mathcal{M}$ is a power of q, $\gcd(q+1, \mathcal{M}) = 1$. Thus $\gcd(2(q+1), \mathcal{M})$ is 1 or 2; in the second case $(q+1)^2 + 1$ is even. Equation (9) therefore defines a soluble congruence class for A, containing arbitrarily large positive representatives. Choose one above L and large enough to make $h_0 \geq 0$. Then supply the standard common-index witnesses at this base. All semantic values were fixed beforehand.

There are eleven modular requests: one exact binomial and ten ranges. Their upper indices are positive and need not be included in W; the indexed-Pell graph certifies those indices directly. Fifteen independent indices suffice for the six exact powers and eleven residues together.

3.5 The explicit numerical source and its finite proof flow

For completeness, the arithmetic source is specified here before applying the inherited calibration. Its constructors are

$$\begin{aligned} I(a,b) &= 2(a+b)(a+b+1) + 4b, & M(a,b) &= I(a,b) + 1, \\ A(a,b) &= I(a,b) + 2, & E(a,b) &= I(a,b) + 3, \\ N_0(a) &= I(a,1), & X_0(a,b) &= N_0(A(a, N_0(b))). \end{aligned}$$

The subscript here distinguishes the formula constructors from the global CRT modulus N and Pell abscissas. Define

$$\begin{aligned}
\mathrm{req}(a,k,m) &= A(a, I(m, N_0(A(a, I(m, N_0(M(a,k)))))))),\\
s(a,b,k) &= A(k, I(M(k,a), M(k,b))),\\
u(a,b,k,m) &= N_0(A(a, I(s(a,b,k), \mathrm{req}(b,k,m)))),\\
\mathrm{clos}(a,b,k,m) &= A(b, I(M(b,k), N_0(A(a, I(m, u(a,b,k,m)))))),\\
v(a,b,c) &= I(M(a,b), A(c, M(a,b))),\\
h(a,b,c) &= A(c, I(\mathrm{clos}(a,b,c,c), N_0(\mathrm{req}(b,c,c)))),\\
\mathrm{bundle}(a,b,c) &= N_0(I(v(a,b,c), h(a,b,c))).
\end{aligned}$$

The nine conclusion polynomials are

$$\begin{aligned}
c_1 &= X_0(a, E(a,b)),\\
c_2 &= I(A(a, I(b,c)), I(A(a,b), A(a,c))),\\
c_3 &= I(E(a,b), I(M(c,a), M(c,b))),\\
c_4 &= I(A(a, A(b,c)), A(b, A(a,c))),\\
c_5 &= I(E(c,a), I(M(c,b), M(a,b))),\\
c_6 &= I(X_0(a, A(a,b)), b),\\
c_7 &= I(E(c,a), I(E(c,b), A(M(a,b), E(a,b)))),\\
c_8 &= I(I(I(c,d),a), I(I(a,c), I(b,c))),\\
c_9 &= \mathrm{bundle}(a,b,c).
\end{aligned}$$

Their code definitions are matched to the retained C222 calculus. The first eight are unconditional axiom families. The ninth is admitted when $c \neq a, b$, represented by $G = (c-a)^2(c-b)^2 - (d+1)^2 = 0$. Write $w(z) = 1 + Fz$. For the eight local columns $(a, b, c, d, S, S', u, v)$ define

$$\begin{aligned}
H &= (S'-S)^2 \prod_{j=1}^{8} (S'w(c_j) - S)^2\\
&\quad \cdot \left((S'w(c_9) - S)^2 + G^2\right)\\
&\quad \cdot \left(S'w(b) - Sw(a)w(I(a,b))\right)^2 \left(S'w(A(a,b)) - Sw(b)\right)^2,\\
\mathcal{L}(i) &= H + \left(x - S - (1 + (i+1)y)u\right)^2 + \left(x - S' - (1 + (i+2)y)v\right)^2,\\
J &= \sum_{j=1}^{9} c_j + a + b + I(a,b) + A(a,b) + 1.
\end{aligned}$$

The local bounds are (t, t, t, t, y, y, x, x), together with the derived condition $J \leq t$. The endpoint equations are

$$x = F + 1 + (y+1)u_0, \qquad y = F + 1 + r_0, \qquad x = 1 + (1 + (t+1)y)u_t.$$

Every factor of H is nonnegative. A zero selects truth, a genuine axiom, modus ponens, or generalization, and gives the appropriate accumulator identity

$$S'w(\mathrm{conclusion}) = Sw(\mathrm{premise}_1)w(\mathrm{premise}_2).$$

Unused premises have truth label 0 and weight one.

Because $F = t!$, the weights $1 + jF$ for $1 \le j \le t$ are pairwise coprime: a prime common to weights $j \ne k$ would divide $j - k$, hence divide F, which is impossible. The beta equations identify overlapping accumulators by uniqueness of remainders bounded by y. Their positive initial value and positive weights propagate positivity. Multiplying the local identities and cancelling gives

$$\prod w(\text{conclusions}) = w(1) \prod w(\text{premises}).$$

Taking valuations at a prime of each positive label's weight gives a multiplicity excess of one at label 1 and zero at every other positive label. If 1 were outside the derivable closure, each selected rule with conclusion outside it would have a premise outside it. Counting these occurrences contradicts that excess. This argument allows cyclic or repeated selected rules; it does not assume the supplied flow is a tree.

Conversely expand a finite refutation to a proof tree, choose the horizon above its length, parameters and all finite J-values, and pad with truth steps. Define $F = t!$ only after this choice. Expanding the pending proof multiset backwards gives the accumulator products. Choose y large and divisible by $t!$, then choose x by CRT for the beta moduli. This is the inherited complete proof-flow construction, with no dependence of the horizon on the later accumulator sizes.

3.6 Reciprocal prime-power decoding and the last carry bounds

With $F = t!$, let

$$d_i = 1 + F^2/i \quad (1 \le i \le t), \qquad B = \prod_{i=1}^{t} d_i = \binom{F^2 + t}{t}, \qquad N = B^K.$$

The d_i are integral, pairwise coprime, and have no prime divisors at most t. These facts follow from $i \mid F$ and $id_i - jd_j = i - j$. Introduce nine positive encoded representatives, the eight local columns A_j and index C, satisfying

$$\begin{gathered} C + \sum_j A_j + g = 9N^2, \qquad B \mid C + F^2 + 1, \\ N \mid \binom{C}{t}, \qquad N \mid \binom{A_j}{b_j + 1}, \qquad N \mid \mathcal{L}(C, A), \qquad N \mid \binom{J(A)}{t+1}. \end{gathered} \tag{13}$$

The b_j are the local bounds. The derived value $J(A)$ is deliberately excluded from the aggregate bound.

The elementary one-sided range lemma states

$$p^e \mid \binom{a}{k} \quad \Longrightarrow \quad a \bmod p^e < k.$$

For $a \ge k$, the p-valuation of the binomial is the sum of $\lfloor a/p^j \rfloor - \lfloor (a-k)/p^j \rfloor - \lfloor k/p^j \rfloor$, each zero or one. If the interval $[a - k + 1, a]$ had no multiple of p^e, all terms from $j = e$ onward would vanish, contradicting valuation at least e. The case $a < k$ is immediate. Its converse is false and is not used.

Choose a prime $p \mid d_i$, write $m = v_p(d_i)$ and $e = Km$. The index congruence and $p > t$ identify $C \bmod p^e = i - 1$. The other ranges give bounded residues for the local columns. A retained literal majorant certificate gives $\mathcal{L}, J < R^E$, where $E = 2^{23}$. Since $R \ge 16! > 2^{44}$, elementary induction gives $R^E < 2^R < 2^K \le p^e$. Polynomial congruence now yields $\mathcal{L}(i - 1, \text{local}) = 0$ and $J(\text{local}) \le t$.

For completeness choose the representatives by CRT at precision d_i^{2K}, so they lie in $[1, B^{2K}] = [1, N^2]$. Their falling factorial numerators have a factor of valuation at least $2Km$; division by $k!$ loses at most

$k-1 \le K-1$. This proves the required divisibilities by p^{Km}, including the derived polynomial range. Precision $2K$ is essential to this argument; no false converse to the range lemma is used.

The remaining digit bounds also remain strict for the new radix $q = N^K$. For $n \le 9N^2$, $1 \le k \le R-1$, and $j \le k$,

$$\binom{n}{j} \le n^k \le 9^k N^{2k} < 2^{4R} N^{2k} \le N^{2k+2} \le N^K = q.$$

For the derived argument the positive-DAG certificate gives $J(a,b,c,d) \le (n+1)^\ell$ when the four inputs are at most n, with $\ell = 2^{22}$. Since $n \le 9N^2$ and $N \ge 10$, $J(A) \le N^{3\ell}$. The finite inequality $2 \cdot 16! \ge 3\ell \cdot 17$ persists by induction, so $3\ell(t+1) \le 2F < K$. Therefore

$$\binom{J(A)}{j} \le J(A)^{t+1} \le N^{3\ell(t+1)} < N^K = q.$$

This last strict inequality is necessary: using the old statement $N^K < q$ would be incorrect for the new radix.

3.7 Closing the theorem and reproducing the finite object

Soundness first recovers the six exact powers from the independent Pell frame. It then uses (5) to interpret the single exact binomial and the factorial ceiling, before invoking any CRT or proof-label meaning. Only afterward are the ten ranges interpreted, the bounded local source decoded, and the Horn-flow theorem applied. The inherited ordinary calibration is

$$\mathrm{C222}(1) \iff \text{finite ZF refutation} \iff \text{finite ZFC refutation}.$$

Its named interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The construction uses the numerical-calculus endpoint before the machine-specific theorem; it does not simulate K215 or instantiate a Jones universal polynomial.

Completeness chooses, in order, the finite proof and padded horizon; the factorial, proof-flow and beta data; B, R, K, N, q and the CRT representatives; the six powers and eleven true binomial residues; and finally a lower Pell frame and sufficiently large A in the common-root congruence class. The common-index theorem supplies all auxiliary Pell witnesses. Every displayed natural quotient is then reconstructed. The checked natural projection gives a zero of P_{319}. Conversely its restoration certificate returns every deleted coordinate, so it preserves both existence directions.

The optional square-shell companion has 320 natural variables, 255 quadratic rows, 13 828 terms and coefficient height 15 248 812. It trades one extra variable and more terms for smaller coefficients, using the retained constructor-preserving syntax transport. The main 319-variable result stays with the original Cantor encoding.

The complete source bundle uses only the Python 3 standard library. Normal and optimized-interpreter replays rebuild both equations and compare all twenty generated JSON files byte for byte. The independent checker verifies every restoration, every final coefficient, all live coordinates and deliberately corrupted controls. Separate finite regressions cover 18 049 factorial cases, 270 independent radix bounds, 49 growth budgets, 1 530 Pell-residue instances and 516 digit and divisibility instances. Those checks are arithmetic regressions, not proofs of unbounded statements or searches for a ZFC refutation.

4 Eleven integer variables with a short divisibility carrier

This section gives the degree-440 equation and two cheaper companions. Its two essential reductions are to replace a fifth-power carrier by a first-power carrier, and to replace a variable-dependent index modulus by $4q^4$. The full supported-modulus interpolation theorem is needed for the converse. Reusing only the older interpolation modulo the strong ordinate would leave a gap.

4.1 The fixed constants and the complete polynomial

The only unknowns are

$$(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}.$$

Use the unchanged source-34 tuple

$$\beta, K_0, M_C, M_F, A_c, B_c, \qquad \beta = 2^{1064921} - 1, \quad A_c = 2^{1113}, \quad B_c = 2^{666104}.$$

The fourth port M_F is the *already shifted* mask. The six exact numerals, their bit lengths, hashes, and executable finite reconstruction are in `norm/coefficients`; their common identity is `mixed-parity-unrounded-1064921`. These are constants, not six extra unknowns or tunable parameters. The tuple must not be mixed with another source's smaller coefficients.

Define the following polynomial abbreviations:

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4,\\
X &= qW, & Y &= (2v + 1)q^3, & a &= (X + 1)Y,\\
H &= 4a + 3, & \Delta &= (a + 2)^2 - 1,\\
F &= (K_0 + W)C - z(q - 1),\\
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
k &= R + 1 + 4hq^4, & \eta &= c - Yk,\\
D_{\mathrm{err}} &= (A_cF - q)^2 + (B_cC - q)^2,\\
\mathcal{B} &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2,\\
\ell &= XY^2, & D_\tau &= 1 + \ell(\ell + 1)k^2,\\
D &= 3\mathcal{B} - 4, & D_g &= 1 + 4DD_\tau s^2,\\
P &= D_\tau D_g,\\
I &= 2rc^2, & D_f &= 1 + \Delta I^2,\\
Q &= \Delta^2 I^2, & \delta &= Q - 1,\\
V &= D_f(c - R + 4IT) - c, & C_y &= QV^2 - 1,\\
D_a &= \delta C_y, & K &= X(3c - 2X) + 2,\\
z_c &= HT + K, & \mathcal{A} &= (z_c^2 + 3H^2)(z_c^3 - 19H^3),\\
S &= srH, & Z &= S(n + (2Q - 1)V) - \mathcal{A},\\
u &= S^2D_f, & b &= 4S^2D_a, & e &= 16S^2P.
\end{aligned}$$

The equation is

$$\boxed{\Big[(Z^2 - u - b - e)^2 - 4(ub + ue + be)\Big]^2 - 64ubeZ^2 = 0.} \tag{14}$$

It contains only fixed nonnegative exponents and integer arithmetic. Its literal division-free evaluator is `norm/scalar11_q4_center.json`: 126 operations and exact degree 440. The following proof uses radicals to analyze the polynomial; no radicals or concealed existential witnesses occur in the equation itself.

The constant recipe constructs 943 fixed tiles from the retained literal 430-action table, assigns the fixed bit weights and positional offsets, and forms the two masks and convolution constant by finite sums. It never simulates the machine. Its code and complete input data are included; the check reconstructs all four native ports and separately checks A_c, B_c. The six uncompressed numeral payloads total 615,883 bytes. Thus a short 126-operation evaluator does not mean that the full coefficient description has only 126 symbols.

4.2 The algebraic norm and the exceptional coordinates

Let

$$\mathcal{F}_8(Z; u, b, e) = [(Z^2 - u - b - e)^2 - 4(ub + ue + be)]^2 - 64ubeZ^2.$$

In a free multiquadratic algebra this is the product of

$$Z - \epsilon_1\sqrt{u} - \epsilon_2\sqrt{b} - \epsilon_3\sqrt{e} \quad (\epsilon_i \in \{-1, 1\}).$$

This identity holds even when radicands coincide or are negative or zero. An independent exact checker multiplies all eight factors and compares all 35 collected coefficients.

The fixed affine front gives q even, $q \equiv 2 \pmod 3$, $|X| \geq 64$, $|Y| \geq 4096$, H odd and nonzero, and $\Delta > 1$. Also $D_\tau \geq 1$ because $\ell(\ell + 1) \geq 0$ for every integer ℓ. The polynomial

$$\pi(x) = (x^2 + 3)(x^3 - 19)$$

has no rational root. It has a root modulo every positive integer, by the retained elementary prime-power and CRT proof in `norm/source/PRIME_TWELVE.md`. Since

$$\mathcal{A} = H^5\pi(T + K/H),$$

we have $\mathcal{A} \neq 0$ for every integer assignment.

If $S = 0$, every scaled radicand in (14) is zero and the polynomial is $\mathcal{A}^8 \neq 0$. Thus $r, s \neq 0$ at a zero. This does *not* yet imply $c \neq 0$. The order of recovery matters. We first prove $D_a > 0$ in both possible cases. If $c = 0$, then $I = Q = 0$ and $D_a = (-1)(-1) = 1$. If $c \neq 0$, then $|I| \geq 2c^2$, $Q > 1$, and $D_f > |c| + 1$. As $V = D_f b_0 - c$ for an integer b_0, either $b_0 = 0$ and $|V| = |c|$, or $|V| \geq D_f - |c| > 1$. Thus $C_y, \delta, D_a > 0$.

In a vanishing conjugate factor, $P = D_\tau D_g$ is therefore the sole possibly negative radicand. If it were negative its nonzero imaginary contribution could not cancel the two real roots and real center. Since D_g is odd, it follows that $D_g > 0$. If $\mathcal{B} \leq 1$, then $D \leq -1$ and $D_g \leq -3$, a contradiction. Hence $\mathcal{B} \geq 2$. For $c = 0$, however,

$$\mathcal{B} = k^2\{q^2(1 - 4Y^2) - D_{\mathrm{err}}\} \leq 0.$$

Consequently $c \neq 0$, and only now do we invoke the strong and auxiliary Pell equations with $I \neq 0$ and $Q > 1$.

4.3 Square recovery and auxiliary integrality

The normalized vanishing factor is a rational signed sum of

$$\sqrt{D_f}, \qquad 2\sqrt{D_a}, \qquad 4\sqrt{P}.$$

All three radicands are positive odd integers. In the squarefree decomposition of their roots, the integer square-root coefficients are odd. Within a nontrivial squareclass the weights have distinct 2-adic orders $0, 1, 2$; the term with least order cannot cancel. Linear independence of distinct squarefree radicals therefore forces all three roots to be integers. Furthermore

$$\gcd(D_\tau, D_g) = \gcd(D_\tau, 1 + 4DD_\tau s^2) = 1,$$

so $D_\tau = \tau^2$ and $D_g = g^2$ separately.

The square root of D_a initially gives a rational y satisfying

$$QV^2 - (Q-1)y^2 = 1. \tag{15}$$

We recall the relevant integrality argument explicitly. In lowest terms $y = a_1/b_1$, one has $b_1^2 \mid Q - 1$. For $|V| > 1$ the usual adjacent-coefficient Pell transformation is

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

After choosing signs, it satisfies $|V'| < |V|$, keeps V' integral, and preserves the reduced denominator of y, since $2Q - 1 \equiv 1 \pmod{b_1}$. The norm excludes $V' = 0$. Descent ends at $|V| = 1$, where $|y| = 1$. Thus $b_1 = 1$. All normalized roots in the factor are now integers, and the factor equation yields $S \mid \mathcal{A}$.

4.4 The smaller index modulus

The retained signed rank proof uses a first-Pell congruence modulo $|XY| > 3R+1$. We may instead use

$$M = 4q^4 > 3R + 1.$$

The inequality follows from the unchanged packing bound $0 < R < q^4 - 2$. Since $4 \mid W$,

$$2XY^2 = 2W(2v+1)^2 q^7 \equiv 0 \pmod{M}.$$

The signed first Pell parameter $P_0 = |2XY^2 + 1|$ is therefore $\operatorname{sign}(X) \pmod{M}$. Its classification gives $|k| = 2\psi_{P_0}(n_0)$ and the congruence

$$\epsilon\, 2n_0 \equiv R + 1 \pmod{M}, \qquad \epsilon = \operatorname{sign}(k)\operatorname{sign}(X)^{n_0 - 1}.$$

For the positive sign $2n_0 \geq R + 1$; for the negative sign $2n_0 \geq M - R - 1 > 2R$. The remaining strong/auxiliary folding argument does not use h separately. It gives a main index $p = R$ with $p > n_0$. Hence the negative branch is impossible. Both $2n_0$ and $R + 1$ lie strictly between zero and M, so

$$2n_0 = R + 1.$$

Only after this equality is proved do we reduce the same recurrence modulo $|XY|$. It restores $k \equiv R+1 \pmod{XY}$ and the old integral coordinate $h_{\text{old}} = (k - R - 1)/(XY)$. The downstream source argument is consequently unchanged. Conversely the canonical positive first-Pell witness has the new congruence modulo M, so $h = (k - R - 1)/M$ is integral. This changes neither the witness domain nor the coefficient tuple.

4.5 Why fifth-power divisibility suffices

The strong and auxiliary norms, the actual residues

$$V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c},$$

the condition $c^2 \mid I$, and the centered budget strip give the pre-projection conclusions of the retained signed folding theorem:

$$R = 2t_0 + 1, \quad t_0 \geq 24, \quad |Y| > \frac{(|X| - 3)^{t_0}}{2} - 1.$$

These conclusions precede the missing main projector; it is not being assumed in order to prove itself. In particular, with $x = |X|$,

$$|H| > x(x-3)^{t_0} > x^{t_0+1}/2^{t_0}, \qquad x \geq 2^{64}.$$

From $S \mid \mathcal{A}$ and $H \mid S$, reduction modulo H gives $H \mid K^5$. Define $K_q = X(X + 2ac) - 1$. The exact identity

$$K = HXc - 2K_q$$

and oddness of H give $H \mid K_q^5$. The signed Pell projection congruence is

$$2^R K_q \equiv \Pi \pmod{H}, \quad \Pi = (\sigma X - 2^R)(2^R \sigma X + 1), \quad \sigma = \operatorname{sign}(c).$$

Thus $H \mid \Pi^5$. Each factor of Π has absolute value smaller than $2^{R+1}x$, and hence

$$|\Pi|^5 < 2^{20t_0+20} x^{10} < |H|.$$

For the last inequality, use

$$64(t_0 - 9) - (21t_0 + 20) = 43t_0 - 596 \geq 436.$$

Therefore $\Pi = 0$, its odd second factor is nonzero, and $\sigma X = 2^R$. This recovers the exact old projector and permits the retained orientation, half-binomial recovery, and source restoration. The invalid inference $H \mid K^5 \Rightarrow H \mid K$ is never used.

4.6 Completeness with the strong pair fixed

Start with the canonical outer tuple from a finite refutation. It has $\mathcal{B} > 64$ and $D_\tau = \tau^2$. Moreover $D_\tau \equiv 1 \pmod 3$, so DD_τ is positive and $2 \pmod 3$, hence a nonsquare. A positive Pell solution, squared once, supplies $s, g > 0$ with

$$g^2 = 1 + 4DD_\tau s^2.$$

Choose one positive strong pair with $4sc^2 \mid I$ and $f^2 - \Delta I^2 = 1$, and set $r = I/(2c^2)$. Then r is an integer divisible by $2s$. Every prime of sr divides I, although sr need not divide I.

The supported-modulus interpolation theorem retained from Round 14 says that, for this *fixed* strong pair, every residue class modulo any positive modulus whose primes divide I occurs as T in an integral auxiliary solution (15). Its proof uses a Pell period $\lambda = 4m_0 f$ and an index increment $d = \operatorname{lcm}(\lambda, 4I)$. For each $p \mid I$, the resulting map $h \mapsto T(h)$ satisfies

$$v_p(T(h_1) - T(h_2)) = v_p(h_1 - h_2).$$

It is therefore a permutation modulo each supported prime power, and CRT supplies the specified modulus. The full proof, including the 2-adic case and the precise initial index, is retained in `norm/source/POWERED_SATURATION_PARENT.md`, Section 5, and was independently reviewed for this use.

The canonical source already has $H \mid K$; write $\kappa = K/H$. Choose a root of $\pi(T+\kappa)$ modulo sr, and interpolate that class while leaving I, f, r fixed. Then

$$\frac{\mathcal{A}}{S} = \frac{H^4 \pi(T+\kappa)}{sr} \in \mathbb{Z}.$$

Finally choose

$$n = \mathcal{A}/S + f + 2\delta y + 4\tau g - (2Q-1)V.$$

One factor of the norm vanishes. The order is outer tuple, guard pair, fixed strong pair, intersective residue, auxiliary solution, and n. This proves completeness without changing a strong witness after a residue has been selected.

4.7 Exact degree and the two cheaper companions

For the primary equation the important degrees are

register	k	$\mathcal{B}$	D_τ	P	I	S	D_f	Q	V	D_a
deg	5	20	30	82	3	8	18	30	23	106.

Thus $\deg(u,b,e) = (34, 122, 98)$. Put $B_0 = Sn - \mathcal{A}$ and $C_0 = S(2Q-1)V$, so $Z = B_0 + C_0$. The universal identity

$$b - C_0^2 = S^2(-V^2 - 4Q + 4)$$

gives $\deg(Z^2 - b) = 96$. Writing $\omega = Z^2 - b$, the inner bracket of (14) is

$$(\omega - u - e)^2 - 4(ub + ue + be).$$

Its unique highest-degree contribution is $-4be$, of degree 220. The correction $64ubeZ^2$ has degree 376. Hence the total degree is exactly 440, with leading form $16b_0^2e_0^2$. On the authentic coefficient slice a specific attained coefficient is

$$[j^{264}w^{40}v^{52}h^{12}s^{12}r^{24}c^{32}z^4]P_{440} = 2^{476}3^{306}\beta^{264} > 0.$$

The exponent sum is 440. The checker independently propagates leading forms through the actual graph and verifies the cancellation identity. An identical 136-operation graph exposes all cancellations and has syntactic degree bound 440; the selected 126-operation graph denotes the same polynomial but has the looser naive syntactic bound 488.

Keeping $k = R + 1 + hXY$ instead gives degree 464 and 125 operations. Removing the center shear as well gives degree 488 and 121 operations. Translation of the unrestricted coordinate n preserves zero existence exactly. The two auxiliary identical graphs of degrees 464 and 440 are audit artifacts, not additional frontier claims.

One charged operation is saved by the exact identity

$$P = D_\tau + D(2D_\tau s)^2 = D_\tau(1 + 4DD_\tau s^2).$$

The short-modulus evaluator also shares $4q^2$ with the budget, computing $4q^4 = (4q^2)q^2$ with one additional multiplication. The actual operation counts for the three selected graphs are

degree	multiplications	additions	subtractions	total
488	77	23	21	121
464	79	24	22	125
440	80	24	22	126.

5 A single product equation with 111 arithmetic operations

The three-root norm can be replaced by one equality between products. The price is a higher total degree. The resulting two equations have eleven unrestricted integer variables, using the same entire source-34 coefficient tuple as Section 4:

first-index modulus	degree	operations
XY	966	111
$4q^4$	882	112.

Both improve the inspected all-root fusion equation of degree 1632 in 114 operations. The degree-440 construction remains the lower-degree alternative at a greater arithmetic cost.

5.1 Definition of the two polynomials

Keep the eleven inputs $(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}$ and the common definitions through $P = D_\tau D_g$ in Section 4. For the 111-operation equation use $k = R + 1 + hXY$; for the 112-operation equation use $k = R + 1 + 4hq^4$. Thus, in both cases,

$$D_\tau = 1 + XY^2(XY^2 + 1)k^2, \qquad D_g = 1 + 4(3\mathcal{B} - 4)D_\tau s^2.$$

Replace the strong and auxiliary definitions by

$$\begin{aligned} A &= a + 2, & D_m &= 1 + \Delta c^2, & U &= rc^2 A D_m, \\ D_f &= 1 + \Delta U^2 P, & Q &= \Delta^2 U^2 P, & \delta &= Q - 1, \\ b_0 &= c - R + 4UPT, & V &= D_f b_0 - c, & C_y &= QV^2 - 1. \end{aligned}$$

Use the same short projector and scalar carrier,

$$\begin{aligned} K &= X(3c - 2X) + 2, & S &= srH, & z_c &= HT + K, \\ \mathcal{A} &= (z_c^2 + 3H^2)(z_c^3 - 19H^3), & Z &= Sn - \mathcal{A}. \end{aligned}$$

The requested equation is simply

$$\boxed{\delta Z^2 - S^2 P D_f C_y = 0.} \tag{16}$$

All abbreviations are polynomials and introduce no extra variables. In the literal evaluator,

$$P = D_\tau + (3\mathcal{B} - 4)(2D_\tau s)^2,$$

The short-index graph reuses $4q^2$ from the budget to form $(4q^2)q^2 = 4q^4$, saving one more multiplication. The product UP is shared between $\Delta U^2 P = (\Delta U)(UP)$ and $b_0 = c - R + 4(UP)T$. The complete graphs are `lowops/product11_degree966.json` and `lowops/product11_degree882.json`. Each is bound to all six original fixed numerals; no coefficient parameter remains free.

Theorem 5.1. *Each polynomial in (16) has an integer zero exactly when the retained native source has a finite refutation. Composing its ordinary source calibration gives*

$$\mathsf{ZFC} \vdash (\textit{no zero of the specified polynomial in } \mathbb{Z}^{11}) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

The exact degrees and arithmetic counts are the two rows displayed above.

5.2 Coprimality recovers the signs as well as the squares

The unconditional affine bounds from the preceding section give $A \neq 0$, $H \neq 0$ odd, $\Delta = A^2 - 1 > 1$, and $X \equiv 0 \pmod 8$. For every integer ℓ, $\ell(\ell+1) \geq 0$. Therefore

$$D_\tau \geq 1, \quad D_\tau \equiv 1 \pmod 8, \quad D_g \equiv 1 \pmod 4, \quad P \equiv 1 \pmod 4.$$

In particular $P \neq 0$. Neither D_f nor δ can vanish, because $\Delta > 1$. Also $C_y \neq 0$: the integer identity $QV^2 = 1$ would force $Q = 1$, which is impossible. Put $D_a = \delta C_y$; it too is nonzero. These statements do not assume that P, Q, or D_f are positive.

If $S = 0$, the polynomial becomes $\delta\mathcal{A}^2$, which is nonzero. Indeed

$$\mathcal{A} = H^5\pi(T + K/H), \qquad \pi(t) = (t^2+3)(t^3-19),$$

and π has no rational root. Thus a zero has $S \neq 0$, so $s, r \neq 0$. We deliberately leave $c = 0$ open until the guard is recovered.

The three integers P, D_f, D_a are pairwise coprime. The first two congruence computations are immediate:

$$D_f \equiv 1 \pmod P, \qquad D_a \equiv 1 \pmod P.$$

For the third, reduction modulo D_f gives

$$Q \equiv -\Delta, \quad \delta \equiv -A^2, \quad V \equiv -c, \quad C_y \equiv -D_m, \quad D_a \equiv A^2 D_m.$$

But $AD_m \mid U$, so $D_f \equiv 1$ modulo both A and D_m. Consequently $\gcd(D_f, D_a) = 1$. All gcds refer to absolute values; the computations hold at negative inputs and at $U = 0$.

Multiply (16) by δ and divide by S^2:

$$\left(\frac{\delta Z}{S}\right)^2 = P D_f D_a. \tag{17}$$

A rational square root of an integer is integral. Pairwise coprimality therefore makes $|P|, |D_f|, |D_a|$ separately squares. Since $P \equiv 1 \pmod 4$, it cannot be a negative odd square. Hence $P > 0$, then $D_f = 1 + \Delta U^2 P > 0$, and then (17) forces $D_a > 0$. There are positive integers

$$p_0^2 = P, \qquad f^2 = D_f, \qquad e_0^2 = D_a.$$

The further identity $\gcd(D_\tau, D_g) = 1$ splits P once more:

$$D_\tau = \tau^2, \qquad D_g = g^2, \qquad p_0 = \tau g.$$

Because $s \neq 0$, the inequality $\mathcal{B} \leq 1$ would imply $D_g \leq 1 - 4D_\tau < 0$. Thus $\mathcal{B} \geq 2$. The centered budget now gives the common source signs and $|c/k - Y| < 1/2$. It also excludes $c = 0$: at that value,

$$\mathcal{B} = k^2(q^2 - D_{\mathrm{err}} - 4q^2Y^2) \leq 0.$$

We have obtained $c \neq 0$ and $U \neq 0$ without an initial assumption about the signs of the saturated radicands.

5.3 Recovering the original Pell witnesses

Define, only within the proof,

$$I = Up_0, \qquad T_{\mathrm{old}} = p_0 T.$$

These are integers, $I \neq 0$, $c^2 \mid I$, and $|I| \geq c^2$. The restored equalities are exact:

$$f^2 = 1 + \Delta I^2, \qquad Q = (\Delta I)^2 > 1,$$
$$4IT_{\mathrm{old}} = 4UPT, \qquad V = f^2(c - R + 4IT_{\mathrm{old}}) - c.$$

Moreover $y = e_0/\delta \in \mathbb{Q}$ satisfies $QV^2 - (Q-1)y^2 = 1$. The adjacent-coefficient descent used in Section 4 makes y integral. Its hypotheses $Q > 1$ and integer V have now been established. Cancellation of the nonzero δ in (16) yields

$$Z^2 = S^2 p_0^2 f^2 y^2,$$

so Z/S is an integer. Hence $S \mid \mathcal{A}$ and $H \mid K^5$.

The complete signed folding and rank argument now has its original hypotheses: the budget strip, the first Pell square, a nonzero strong ordinate divisible by c^2, and the auxiliary norm with $V \equiv -c \pmod f$, $V \equiv -R \pmod c$. Its use does not require an odd strong multiplier. In the $4q^4$ version the first-index argument is precisely the one already proved in Section 4. The exact index and orientation later restore $XY \mid k - R - 1$, so the original quotient h can be regenerated.

The fifth-power height argument also applies unchanged. With $R = 2t + 1$, $t \geq 24$, $x = |X| \geq 2^{64}$ and

$$\Pi = (\operatorname{sign}(c)X - 2^R)(2^R \operatorname{sign}(c)X + 1),$$

it gives $H \mid \Pi^5$ and $|\Pi|^5 < |H|$. Thus $\Pi = 0$, the exact main projection is restored, and the signed half-binomial argument recovers every outer native condition. Fresh private witnesses supply an actual zero of the retained source. This proves the forward finite-refutation implication using the same source calibration.

5.4 Completeness and the extra interpolation factor

Start with a canonical outer source witness. It has $A > 1$ even, $R \equiv 3 \pmod 4$, $c = \psi_A(R) > 0$, $c \equiv R \pmod 4$, $D_\tau = \tau^2$, and the stronger budget $\mathcal{B} \geq 64$. The positive nonsquare $(3\mathcal{B} - 4)D_\tau \equiv 2 \pmod 3$ has a Pell unit. Squaring one supplies positive s, g with $g^2 = 1 + 4(3\mathcal{B} - 4)D_\tau s^2$. Fix them and put $p_0 = \tau g$.

Choose one positive strong pair I, f satisfying

$$4sAD_m p_0 c^2 \mid I, \qquad f^2 - \Delta I^2 = 1.$$

Set $r = I/(AD_m p_0 c^2)$ and $U = I/p_0$. The strong pair is now fixed. Every prime factor of $M = p_0 sr$ divides I. The canonical projector gives $\kappa = K/H \in \mathbb{Z}$. Choose an intersective residue a_0 modulo sr with $\pi(a_0 + \kappa) \equiv 0 \pmod{sr}$. Supported-modulus interpolation, at this same fixed I, f, supplies the full auxiliary witness with

$$T_{\mathrm{old}} \equiv p_0 a_0 \pmod{p_0 sr}.$$

The extra factor p_0 in this modulus is essential. It makes $T = T_{\mathrm{old}}/p_0$ an integer and ensures $T \equiv a_0 \pmod{sr}$. Therefore

$$\frac{\mathcal{A}}{S} = \frac{H^4 \pi(T + \kappa)}{sr} \in \mathbb{Z}.$$

Finally set $n = \mathcal{A}/S + p_0 f y$. Equation (16) vanishes. The order of choices is outer witness, s, g, one strong I, f, r, a residue, T_{old}, T, y, and n. No coprimality between these moduli is assumed, and the strong pair is never changed after the residue request. This proves the reverse implication.

5.5 Exact degree and finite validation

The key degrees in the two literal graphs are

	k	P	U	D_f	Q	b_0	V	C_y	S	$\mathcal{A}$
XY	7	94	23	152	164	118	270	704	8	35
$4q^4$	5	82	23	140	152	106	246	644	8	35.

In both cases $\deg Z = 35$. The summand δZ^2 has degree 234 or 222, while $S^2PD_fC_y$ has degree 966 or 882. The latter therefore determines the full leading homogeneous part.

Put $q_0 = 6\beta j$, $X_0 = 12q_0w$, $Y_0 = 2vq_0^3$, $a_0 = X_0Y_0$, and

$$k_0 = ha_0 \quad \text{or} \quad k_0 = 4hq_0^4 - q_0^3(12wC - q_0z),$$

respectively. Direct exact multiplication gives the leading form

$$2^{36}3^7q_0^{14}X_0^{70}Y_0^{112}k_0^{42}s^{16}r^{12}c^{40}T^2.$$

For the native modulus its coefficient at $j^{588}w^{112}v^{154}h^{42}s^{16}r^{12}c^{40}T^2$ is $2^{1002}3^{707}\beta^{588} > 0$. For the short modulus the coefficient at $j^{588}w^{70}v^{112}h^{42}s^{16}r^{12}c^{40}T^2$ is $2^{960}3^{665}\beta^{588} > 0$. Their exponent sums give the two stated exact degrees. The latter coefficient is selected by the maximal h power, so no cancellation from the other leading summands is possible.

The independent checker uses only the Python standard library and does not import the producer. It verifies every local polynomial formula, all 135 or 136 live nodes, all six complete coefficient payloads, the coprimality remainder identities, both saturation identities, the new index-modulus identity, and exact leading monomials at every gate. Five altered mathematical formulas in each graph are rejected. Normal and optimized Python runs agree, and both graphs rebuild byte-for-byte. These finite checks certify the exported algebra and counts; the unbounded semantic argument is the proof above and its explicitly retained source lemmas.

6 Reproduction, validation, and remaining work

6.1 What the archive contains

The root `README.md` is the execution guide. The compiled PDF and all its editable TeX sections are included. The mathematical artifacts are organized as follows.

`direct/`
: The literal 319-variable quartic, all 254 quadratic residuals, its fully collected 10,283-term coefficient list, readable versions of both lists, the raw source, and every natural-valued restoration polynomial. The builder, independent verifier, source theorem, imported compiler code, original notices, source ledger, and comparison receipt are supplied. The optional `equation320/` companion uses 320 naturals and 255 residuals, has 13,828 terms and height 15,248,812; it trades one variable and more terms for much smaller height through the inherited square-shell syntax translation.

`norm/`
: The three selected eleven-integer DAGs of degrees 488, 464, and 440; two polynomial-identical cancellation-exposing audit DAGs; the six literal coefficient numerals and their reconstruction recipe; all retained ordinary proof dependencies; and the new exact finite checker.

`lowops/`
: The two product-form eleven-integer equations of degrees 966 and 882, their ordinary proof, literal graphs, independent polynomial and degree audit, and source bindings to the same fixed tuple.

`reviews/`
: Separate mathematical reviews of the three new transformations. These are independent research-agent audits within this workflow, not an external human endorsement. Their scope is stated in each file.

The top-level SHA-256 manifest covers the shipped files. The hashes are integrity checks, not substitutes for the actual numeral data or proofs.

6.2 The direct finite audit

The principal raw system has 465 natural coordinates and 400 quadratic rows. The delivered contraction removes 146 coordinates and 146 rows. For each deletion the certificate gives a polynomial restoration that is nonnegative on the remaining natural coordinates. The independent checker restores each deleted row exactly, verifies every substitution, expands the entire sum of squares, and compares every collected coefficient. This proves a finite equation-to-equation equivalence independently of the number-theoretic interpretation of the raw system.

The readable residual and polynomial files are parsed back into the same coefficients. Altered residual, restoration, and coefficient controls are rejected. Both normal and optimized Python execution rebuild the principal and companion artifacts, comparing all 20 generated JSON files byte for byte in temporary directories. The complete 349-variable baseline is also re-expanded and its contraction replayed, so the table does not rely only on an inherited headline count.

The semantic regression suite checks 18,049 factorial cases, 270 radix bounds, 49 growth budgets, 1,530 modular Pell residue instances, and 516 digit/divisibility instances. These finite cases are useful implementation checks. The unbounded lemmas are proved in the report and retained source proofs; their validity is not inferred from these sample counts.

6.3 The small-variable finite audits

The norm checker does not import its producer. It binds every live register to an independently transcribed formula in a free polynomial ring, checks all 35 coefficients of the three-root norm, proves the center cancellation and short-projector identities, checks topology and liveness, and propagates an exact nonzero leading coefficient through each actual graph. The primary degree-440 coefficient displayed in Section 4 is therefore an attained coefficient on the actual fixed-coefficient slice.

The product-form checker likewise independently binds the complete register definitions, verifies the pairwise-coprimality congruences and the exact leading induction, and counts every live operation. A separate review rederived its leading forms and confirmed the final 111/112-operation census. Reusing the existing $4q^2$ register saves one operation in the second graph without changing its polynomial.

All new checks use explicit exceptions rather than removable Python assertions. The inherited coefficient constructor deliberately requires normal execution; the normal audit rebuilds it and compares every native coefficient numerically with the authenticated payload. Optimized checks retain the same payload authentication and all new exact algebra checks. The archive does not bypass that constructor's execution contract.

6.4 Offline commands

Only Python 3's standard library is needed for the polynomial artifacts. The root reproduction wrapper supplies the exact command set; the main component commands are:

```
python3 -B direct/verify.py
python3 -B direct/check_semantics.py
python3 -B direct/reproduce.py
python3 -B -O direct/reproduce.py
python3 -B norm/check.py
python3 -B -O norm/check.py
python3 -B lowops/check.py
python3 -B -O lowops/check.py
```

No command searches for a root or decides consistency. The TeX source can be compiled with a standard TeX Live installation, running `pdflatex -interaction=nonstopmode -halt-on-error report.tex` in `report/` until references stabilize.

6.5 What remains open

The strongest unresolved issue is a genuine lower bound or a new architecture that improves the small-variable and small-coefficient objectives together. This work does not reduce the minimum variable count already achieved elsewhere in the repository. It improves the eleven-integer degree/circuit tradeoffs and a direct modest-height quartic architecture.

Two next steps are concrete. First, internalize the new signed product recovery and supported-modulus witness construction in the proof assistant, keeping its calibration theorem explicit. Second, investigate whether the shared factorial/CRT certificate admits a further reduction of the independent index certificates or bounded local columns. Any such claim must preserve the noncircular digit proof and the complete refutation witness order. Merely hiding a certificate inside an unexpanded abbreviation does not reduce its charged complexity.

References

[1] V. Reshetnikov and ProveIt contributors, *ProveIt*, pinned commit `45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt. The precise parent proof texts, numerical payloads, and source notices used here are retained in the archive.

[2] ProveIt incoming research packets, *Coprime Projection*, *Powered Saturation and Factorial Carriers, Round 14*, *Synchronized Exponents*, and the other incoming packets listed in the provenance inventory, 24 October 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6/docs/incoming.

[3] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* **47** (1982), 549–571. https://doi.org/10.2307/2273588.

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[5] J. Bayer, M. David, M. Hassler, Yu. Matiyasevich, and D. Schleicher, Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization, arXiv:2506.20909. https://arxiv.org/abs/2506.20909.

[6] M. David and collaborators, Universal Pairs for Diophantine Equations, *Archive of Formal Proofs*, 2025. https://isa-afp.org/entries/Diophantine_Universal_Pairs.html.